PAPER SUMMARY

AI-Driven Identity Verification: A Continuous Trust Architecture for Global Payment Platforms

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ABSTRACT

Identity verification in digital payment platforms is almost universally implemented as a one-time onboarding event. This design assumption breaks down in practice: fraud actors exploit the static trust boundary using behavioral drift, synthetic identity construction, and coordinated low-and-slow account manipulation that evades snapshot-based detection entirely. This paper presents a continuous trust architecture that reframes identity verification as an ongoing probabilistic process rather than a discrete checkpoint. The architecture integrates behavioral telemetry, device intelligence, and transaction context into a layered evaluation pipeline anchored by a temporal trust propagation model using Isolation Forest-based anomaly detection and a dynamic trust scoring engine. Analytical evaluation using the trust propagation model $T_t = \alpha T_{t-1} + (1-\alpha)P(I|X_t)$ demonstrates how the system differentiates between legitimate users, gradual behavioral drift, and sudden account compromise, while sensitivity analysis of α characterizes the stability-responsiveness tradeoff. The architecture incorporates governance mechanisms for auditability, explainability, and regulatory compliance, and is designed for deployment in high-throughput distributed payment environments.

HIGH-LEVEL ARCHITECTURE

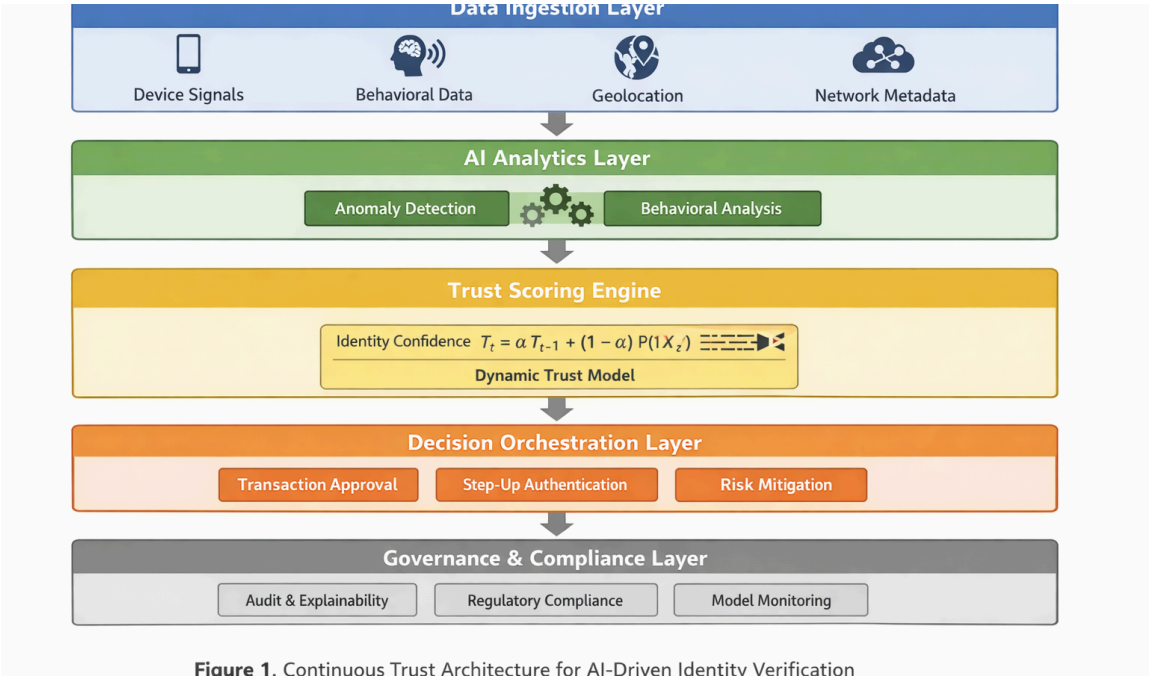


Figure 1. Continuous Trust Architecture for AI-Driven Identity Verification

Fig. 1. Continuous trust architecture for AI-driven identity verification in global payment platforms.

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